
NOW OPEN-SOURCE

NEXUS

An enterprise AI framework

for secure, grounded, context-aware intelligence.

github.com/<your-org>/nexus

Swipe →

Data-rich. Insight-poor.

Enterprises sit on terabytes of fragmented data, yet most AI prototypes never reach production. Teams reinvent ingestion, retrieval, guardrails, RBAC, and audit logging for every new use case.

7 layers. One front door.

A layered Python framework built so each layer is independently installable, replaceable, and deployable. The root nexus package is the only entry point and never imports child-layer code.

Library

from nexus import ...

CLI

\$ nexus ...

REST API

POST /v1/ask

Loose coupling, end to end.

1

Enterprise Data Pipeline

API · batch · stream · CDC

2

Processing & Enrichment

ETL · chunking · metadata

3

Embedding & Retrieval

vector · lexical · hybrid · graph

4

Orchestration & Guardrails

PII · prompt safety · grounded RAG

5

Experience API

REST · SDK · assistant · web · mobile

6

Security & Governance

RBAC · tenant isolation · AEAD · audit

7

Observability

metrics · logs · traces · alerts

Capabilities

- Connectors for REST, batch, streaming, CDC
- Document chunking + metadata enrichment
- Hybrid vector + lexical + graph retrieval
- Grounded RAG with citations and confidence
- PII masking, prompt safety, policy engine
- REST API, Python SDK, CLI, assistant channels
- RBAC, tenant isolation, authenticated encryption
- Metrics, logs, traces, AI events, alerts

Hardened by default.

- Fernet AEAD (AES-128-CBC + HMAC-SHA256), fail-closed key handling
- API-key auth with constant-time compare; pluggable RBAC hook
- Session ownership enforced; spoofable user_id removed
- SSRF and bearer-token leak defense in API connectors
- Subprocess argv + path-traversal hardened end to end
- NFKC-normalized guardrails; Luhn-validated PII

Install in seconds.

```
$ pip install -e nexus
$ nexus validate-platform configs/nexus.yaml
$ nexus prepare-demo configs/nexus.yaml
$ nexus ask configs/nexus.yaml "What is MFA?"
```

```
from nexus import NexusPlatform
platform = NexusPlatform.from_config(path)
print(platform.ask("What is MFA?"))
```

Why teams adopt it.

- Drop-in Python library — no side effects at import
- Loose coupling — swap any layer without touching others
- Security built in, not bolted on
- Deterministic local dev — no required cloud services
- 181 tests across 8 suites, deterministic and offline
- Vendor-neutral — wire in your own LLM, vector DB, KMS, SIEM

GET STARTED

Try Nexus.

Read the full integrator guide:

docs/USING_NEXUS.md

Star ★

Fork

Contribute

github.com/<your-org>/nexus
